Track 1 pipeline report

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Track 1 pipeline report

Team fulopjoz. MVA Hackathon 2026, Track 1, variant prioritization. Submission file: `submission_track1.csv`.

Result

The pipeline ranks the clinically confirmed genotype first of 6,962 candidates: compound-heterozygous **BUB1B p.Leu737Ter** (chr15:40,209,701 T>G, stop gained; ClinVar allele ID 529272) plus **BUB1B p.Asn1002Lys** (chr15:40,220,612 T>G, missense), score 0.5823. Score at rank 2 is 0.5243. The score column `epcr` — the challenge-defined (0, 1] score slot ("estimated probability of causal relationship") — is populated by the pipeline's composite weighting. It is a bounded heuristic rank score, **not an empirically calibrated probability** (no calibration validation exists for this score calibration label; see Section *Method*). Pairs listed are phase-unknown *candidates* — presence of two compound alleles in the same gene does not by itself prove that they are in trans.

Rank	Genotype	Score
1	BUB1B p.Leu737Ter + p.Asn1002Lys	0.5823
2	RAI1 frameshift/frameshift pair	0.5243
3-6	RAI1 missense/frameshift pairs	0.4914
7-10	HLA-DRB1 founder pairs	0.4509
2,212 and below	homozygous-recessive singles (ordered after biallelic pairs by design)	—

About the ordering, explicitly. The two strata are ranked by the same score, but a candidate *pair* aggregates the evidence of two alleles and is the target entity of a biallelic-recessive retrieval task, so the output lists every pair before any homozygous single. That is the declared design, and it matters here: the highest-scoring singles in this run reach 0.6119, above the pair's 0.5823. Under a single global sort by raw score the true pair would rank **8**, not 1 — the interim global-order run measured exactly that and is recorded in the correction ledger; the ranking above is the pair-first declaration that the method text has always carried.

Note: the September 17 table showed three KIF7 rows at ranks 8-10; those were produced by the minus-strand annotation error corrected on 2026-09-29 (the "stop" became p.S1016S synonymous), and the September 29 corrected table already removed them. The table above is the definitive 2026-10-19 run (splice-classification and multiallelic-allele handling audits included).

Method

One pass over the 5,012,204-record proband VCF. Each stage's counter is measured from the run.

1. **Quality filter.** FILTER is PASS or absent, genotype quality at least 20, read depth at least 10, alternate-allele fraction of the called allele at least 0.25. Multiallelic records are evaluated per genotype-called allele (the September run indexed AD[1] regardless of GT). Keeps 4,592,493 records (September's gate kept 4,592,140; the 353-record difference is multiallelic records whose ALT1 failed the fraction gate while a called ALT2 passed).

2. **Effect calling.** Variants are annotated against local GENCODE v47 MANE transcripts. Stop gained, frameshift, and canonical splice donor/acceptor variants count as loss of function (LoF); missense and in-frame indels are scored separately; canonical intronic splice sites are classified strand-aware (the earlier build overwrote them with `splice_region`, which could not seed pairs). Keeps 12,952 coding or splice candidates (the pre-correction carry was 16,963: phantom minus-strand codon mismatches + splice LOF re-classification + multiallelic decomposition; see `docs/CORRECTION_2026-09-29.md`).
3. **Rarity.** Allele frequency from local gnomAD v4.1 exome sites; every record at a multi-record position is merged and alleles match exactly on (chrom, position, ref, alt). ClinVar joins as background evidence only.
4. **Candidate pairing.** Compound-heterozygous pairs (an LoF allele plus a second damaging allele in the same gene; phase unknown – see the disclosure on trans below) and homozygous damaging candidates (missense, stop-gain, in-frame indel). There is no allele-frequency gate at candidate collection: rarity enters only through the score (step 5). Finds 2,211 pairs and 4,751 homozygous candidates (the emitted `hom_singles` counter equals the written rows; the reversed-duplicate inflation that produced 2,372 rows is deduplicated with a deterministic orientation).
5. **Scoring.** Each variant scores as a weighted mix: founder effect 0.45, predicted pathogenicity 0.60, rarity 0.30, plus phenotype similarity to the eight case HPO terms, weight 2.0. A pair scores as the stronger allele × (0.5 + 0.5 × the weaker).

External cross-check

Exomiser 14.0.0 on the same VCF ranks BUB1B first (p = 0.0001) with the same eight HPO terms; without phenotype input its top is a 46-way tie that still contains BUB1B. The records of both runs (`exomiser_summary.tsv` on the analysis host, artifact hashes verified 2026-09-17) are traceable; the answer is therefore not an artifact of this pipeline's own code.

The phenotype weight's influence is characterised the same way: at weight 0 the true pair ranks 14 of 6,962 (epcr 0.6913); at the shipped weight 2.0 it ranks 1 (record: `desensitization_audit19b.json` on the analysis host, measured 2026-10-19 on the definitive table, after the 2026-09-18 record measured 25 of 8,666 on the pre-corrected table and the subsequent sensitivity run measured 21 of 6,921 on the minus-strand-corrected table). We disclose this dependence rather than hide it.

Reproduction

Python 3.12 with pysam, cyvcf2, pyyaml. References (paths in `pipeline/conf/track1_config.yml`): GENCODE v47 MANE, GRCh38 FASTA, gnomAD v4.1 exome sites, ClinVar, HPO with the OMIM gene-to-phenotype map. Pathogenicity comes from consequence class; CADD v1.7 scores are used when local tiles cover a site, and none did in this run, so the CADD term is neutral and all CADD fields in the ranked table are empty.

```
python3 pipeline/scripts/prioritize.py \
    --conf pipeline/conf/track1_config.yml \
    --with-af \
    --with-hpo \
    --out output/annot_full.tsv
python3 pipeline/scripts/make_submission_csv.py \
    --in output/annot_full.tsv \
    --out submission_track1.csv
```

The annotate and HPO modules carry built-in self-tests (`python3 pipeline/scripts/annotate.py`, `python3 pipeline/scripts/_hpo.py`); they print SKIP with the missing reference named when a host lacks the reference files.

Disclosure

Built with AI assistance (Kimi K3 via Oh My Pi); no patient data left the analysis hosts and none was sent to any external API, per the challenge rules.

Data availability

Four artefacts cited above and in the technical documentation are **not present in this repository, by design**. They are pipeline outputs computed from the patient's sequence data, so each one is a per-variant table for a single identifiable individual. Publishing them would disclose case data that this project's whole architecture exists to contain, and the repository becomes public after the hackathon closes.

Cited artefact	Why it is absent
`desensitization.json`	Per-variant ranks for the case under varied phenotype weighting. Patient-derived.
`exomiser_summary.tsv`	Per-variant Exomiser output for the case. Patient-derived.
`output/annot_full.tsv`	The full annotated variant table for the case. Patient-derived.
`annot_ns.tsv`	The nonsynonymous subset of the same table. Patient-derived.

They were used to produce the numbers reported above and are retained on the whitelisted analysis machine. They can be made available to the organisers or to a reviewer under the data agreement that governs the source dataset, on request.

`hpo_gene_to_phenotype.tsv` is also referenced and is likewise not shipped, for a different and more ordinary reason: it is a public reference file distributed by the Human Phenotype Ontology project and is fetched rather than vendored.

Everything needed to re-run the pipeline is here: the scripts under `pipeline/`, the configuration under `pipeline/conf/`, and the reproduction steps above. What is withheld is the output computed from one child's genome, not the method that produced it.